

Causal Machine Learning for Moderation Effects

Nora Bearth and Michael Lechner

University of Copenhagen

Problem

Suppose that the effect of a particular training program for the unemployed is larger for women than for men. By comparing the average treatment effects for these two groups, without taking into account the different distribution of other covariates of men and women, such as education or labor market experience, we might implicitly compare a group with more extended labor market experience, with a group with shorter labor market experience.

Contribution

- Proposing the new parameter BGATE
- Showing three ways to estimate the BGATE

Definitions

N iid observations $H_i = (D_i, Y_i, Z_i, X_i)$ (H_i has N rows so what does i mean?), where the treatment D_i and moderator variable Z_i are binary. A set of $k \in \{1, \dots, p\}$ covariates $X_{i,k}$ where $X_i = (X_{i,1}, \dots, X_{i,p})$. Define potential outcomes as (Y_i^0, Y_i^1) , and potential covariates and potential moderators are defined as $(X_i^0, X_i^1, Z_i^0, Z_i^1)$.

ITE, ATE, CATE, IATE, GATE

- ITE: $\xi_i = Y_i^1 - Y_i^0$
- ATE: $\theta = E[Y_i^1 - Y_i^0]$
- CATE: $\theta^C = E[Y_i^1 - Y_i^0 | X_i = x]$
- IATE: $\tau(x, z) = E[Y_i^1 - Y_i^0 | Z_i = z, X_i = x]$
- GATE: $\theta^G = E[Y_i^1 - Y_i^0 | Z_i = z] = E[\tau(X_i, z) | Z_i = z]$

Δ GATE

If we want the difference in treatment effect between the two groups, we can calculate the Δ GATE:

$$\begin{aligned}\theta^{\Delta G} &= E[Y_i^1 - Y_i^0 | Z_i = 1] - E[Y_i^1 - Y_i^0 | Z_i = 0] \\ &= E[\tau(X_i, 1) | Z_i = 1] - E[\tau(X_i, 0) | Z_i = 0],\end{aligned}$$

which may be difficult to interpret as the two groups may differ in the distribution of other covariates X_i .

Δ GATE

If we want the difference in treatment effect between the two groups, we can calculate the Δ GATE:

$$\begin{aligned}\theta^{\Delta G} &= E[Y_i^1 - Y_i^0 | Z_i = 1] - E[Y_i^1 - Y_i^0 | Z_i = 0] \\ &= E[\tau(X_i, 1) | Z_i = 1] - E[\tau(X_i, 0) | Z_i = 0],\end{aligned}$$

which may difficult to interpret as the two groups may differ in the distribution of other covariates X_i .

Note: This is later in the paper written as

$$\theta^{\Delta G} = E[E[Y_i^1 - Y_i^0 | Z_i = 1] - E[Y_i^1 - Y_i^0 | Z_i = 0]]$$

BGATE and Δ BGATE

The balanced group average treatment effect (BGATE) is designed to balance the distribution of other variables within the groups we want to compare to each other, i.e. $Z_i = 0$ and $Z_i = 1$. Use $W_i \subseteq X_i$ to balance the GATEs and define the BGATE as:

$$\begin{aligned}\theta^B(z) &= E[E[Y_i^1 - Y_i^0 | Z_i = z, W_i]]? \\ &= E[E[\tau(X_i, z) | Z_i = z, W_i]],\end{aligned}$$

BGATE and Δ BGATE

The balanced group average treatment effect (BGATE) is designed to balance the distribution of other variables within the groups we want to compare to each other, i.e. $Z_i = 0$ and $Z_i = 1$. Use $W_i \subseteq X_i$ to balance the GATEs and define the BGATE as:

$$\begin{aligned}\theta^B(z) &= E[E[Y_i^1 - Y_i^0 | Z_i = z, W_i]]? \\ &= E[E[\tau(X_i, z) | Z_i = z, W_i]],\end{aligned}$$

and its difference (Δ BGATE) as

$$\begin{aligned}\theta^{\Delta B} &= E[E[Y_i^1 - Y_i^0 | Z_i = 1, W_i] - E[Y_i^1 - Y_i^0 | Z_i = 0, W_i]] \\ &= E[E[\tau(X_i, 1) | Z_i = 1, W_i] - E[\tau(X_i, 0) | Z_i = 0, W_i]]\end{aligned}$$

Decomposition of Δ GATE

$$\begin{aligned} & E[Y_i^1 - Y_i^0 | Z_i = 1] - E[Y_i^1 - Y_i^0 | Z_i = 0] \\ &= E[E[Y_i^1 - Y_i^0 | Z_i = 1, W_i] - E[Y_i^1 - Y_i^0 | Z_i = 0, W_i]] \\ &+ \frac{P(Z_i = 0)}{P(Z_i = 1)} E[E[Y_i^1 - Y_i^0 | Z_i = 1, W_i]] - E[E[Y_i^1 - Y_i^0 | Z_i = 1, W_i] | Z_i = 0] \\ &- \frac{P(Z_i = 1)}{P(Z_i = 0)} E[E[Y_i^1 - Y_i^0 | Z_i = 0, W_i]] - E[E[Y_i^1 - Y_i^0 | Z_i = 0, W_i] | Z_i = 1] \end{aligned}$$

Assumptions 1: Identifying assumptions

- Conditional Independence: $(Y_i^1, Y_i^0) \perp D_i | X_i = x, Z_i = z$,
 $\forall x \in \mathcal{X}, \forall z \in \mathcal{Z}$.
- Common support: $0 < P(D_i = d | X_i = x, Z_i = z) < 1$,
 $\forall d \in \{0, 1\}, \forall x \in \mathcal{X}, \forall z \in \mathcal{Z}$.
- Exogeneity of confounder: $X_i^0 = X_i^1, Z_i^0 = Z_i^1$.
- Stable Unit Treatment Value Assumption:
 $Y_i = D_i Y_i^1 + (1 - D_i) Y_i^0$.

Lemma 1

Under Assumptions 1 the parameter

$$\theta^B(z) = E[E[Y_i^1 - Y_i^0 | Z_i = z, W_i]]$$

is identified as

$$E[E[\mu_1(Z_i, X_i) - \mu_0(Z_i, X_i) | Z_i = z, W_i]]$$

with $\mu_d(z, x) = E[Y_i | D_i = d, Z_i = z, X_i = x]$.

Lemma 1

Under Assumptions 1 the parameter

$$\theta^B(z) = E[E[Y_i^1 - Y_i^0 | Z_i = z, W_i]]$$

is identified as

$$E[E[\mu_1(Z_i, X_i) - \mu_0(Z_i, X_i) | Z_i = z, W_i]]$$

with $\mu_d(z, x) = E[Y_i | D_i = d, Z_i = z, X_i = x]$.

This means the parameter

$$\theta^{\Delta B} = E[E[Y_i^1 - Y_i^0 | Z_i = 1, W_i] - E[Y_i^1 - Y_i^0 | Z_i = 0, W_i]]$$

is identified as

$$\begin{aligned} &E[E[\mu_1(Z_i, X_i) - \mu_0(Z_i, X_i) | Z_i = 1, W_i] \\ &\quad - E[\mu_1(Z_i, X_i) - \mu_0(Z_i, X_i) | Z_i = 0, W_i]] \end{aligned}$$

Estimation strategy 1: Double/Debiased Machine Learning

Similar to Kennedy (2023) estimation CATE's with the DR-learner:

- Step 1: Estimate doubly robust score function
- Step 2: Regres score function on Z_i and W_i
- Step 3: Take the difference between the two groups defined by Z and average over variables in W .

Score function

Score function is given by

$$\begin{aligned}\hat{\phi}^{\Delta B}(h; \theta^{\Delta B}, \hat{\eta}) = & \hat{g}_1(w) - \hat{g}_0(w) + \frac{z \left(\hat{\delta}(h) - \hat{g}_1(w) \right)}{\hat{\lambda}_1(w)} \\ & - \frac{(1-z) \left(\hat{\delta}(h) - \hat{g}_0(w) \right)}{1 - \hat{\lambda}_1(w)} - \theta^{\Delta B}\end{aligned}$$

where

$$\delta(h) = \mu_1(z, x) - \mu_0(z, x) + \frac{d(y - \mu_1(z, x))}{\pi_1(z, x)} - \frac{(1-d)(y - \mu_0(z, x))}{1 - \pi_1(z, x)},$$

$$\lambda_z(w) = P(Z_i = z | W_i = w),$$

$$g_z(w) = E[\delta(H_i) | Z_i = z, W_i = w],$$

$$\mu_d(z, x) = E[Y_i | D_i = d, Z_i = z, X_i = x],$$

$$\pi_d(z, x) = P(D_i = d | Z_i = z, X_i = x).$$

Theorem 1

Under assumptions 2-6, the proposed estimation strategy for the Δ BGATE obeys $\sqrt{N}(\hat{\theta}^{\Delta B} - \theta^{\Delta B}) \xrightarrow{d} N(0, V^*)$ with $V^* = E[\phi^{\Delta B}(H_i; \theta^{\Delta B}, \eta)^2]$, i.e. the Δ BGATE is $\sqrt{N}$ consistent and asymptotically normal.

Estimation strategy 2: Automatic Debiased Machine Learning

Same three steps as strategy 1 but use Riesz representers instead of propensity scores.

Estimation strategy 2: Automatic Debiased Machine Learning

Same three steps as strategy 1 but use Riesz representers instead of propensity scores.

New score function:

$$\begin{aligned}\hat{\phi}_{Riesz}^{\Delta B}(h; \theta^{\Delta B}, \hat{\eta}) = & \hat{g}_1(w) - \hat{g}_0(w) + \hat{\alpha}_z(w) \left(\hat{\delta}(h) - \hat{g}_z(w) \right) \\ & - \theta_{Riesz}^{\Delta B}\end{aligned}$$

with

$$\delta(h) = \mu_1(z, x) - \mu_0(z, x) + \alpha_d(z, x)(y - \mu_d(z, x)),$$

$$g_z(w) = E[\delta(H_i) | Z_i = z, W_i = w],$$

$$\mu_d(z, x) = E[Y_i | D_i = d, Z_i = z, X_i = x],$$

Estimation strategy 2: Automatic Debiased Machine Learning

Same three steps as strategy 1 but use Riesz representers instead of propensity scores.

New score function:

$$\begin{aligned}\hat{\phi}_{Riesz}^{\Delta B}(h; \theta^{\Delta B}, \hat{\eta}) = & \hat{g}_1(w) - \hat{g}_0(w) + \hat{\alpha}_z(w) \left(\hat{\delta}(h) - \hat{g}_z(w) \right) \\ & - \theta_{Riesz}^{\Delta B}\end{aligned}$$

with

$$\delta(h) = \mu_1(z, x) - \mu_0(z, x) + \alpha_d(z, x)(y - \mu_d(z, x)),$$

$$g_z(w) = E[\delta(H_i) | Z_i = z, W_i = w],$$

$$\mu_d(z, x) = E[Y_i | D_i = d, Z_i = z, X_i = x],$$

No Theoretical guarantees.

Estimation Strategy 3: Reweighting Approach

Idea: Change the data such that the distribution of the balancing variables are the same across the groups defined by the moderator variable. Having balanced data we can estimate the BGATE the same way as the GATE, as they are equal in balanced data.

Estimation Strategy 3: Reweighting Approach

Idea: Change the data such that the distribution of the balancing variables are the same across the groups defined by the moderator variable. Having balanced data we can estimate the BGATE the same way as the GATE, as they are equal in balanced data.

- Step 1: Duplicate each observation in $h_i = \{x_i, z_i, d_i, y_i\}$ such that we create $h_{i,1}^{balanced}$ and $h_{i,0}^{balanced}$. Set z values in $h_{i,1}^{balanced}$ to 1 and in $h_{i,0}^{balanced}$ to 0. Then merge $h_{i,1}^{balanced}$ and $h_{i,0}^{balanced}$ to create $h_i^{balanced}$ that will have $2 \times N$ rows.

Step 2

Next step varies between the paper and the appendix

- For each observation in $h_i^{balanced}$ the covariate values for the balancing variables W_i are compared with those of all other observations that share the same treatment/moderator assignment from the original data.

Step 2

Next step varies between the paper and the appendix

- For each observation in $h_i^{balanced}$ the covariate values for the balancing variables W_i are compared with those of all other observations that share the same treatment/moderator assignment from the original data.
- Using nearest neighbors (Mahalanobis distance metric) each observation is then matched to the nearest observation in the original data.

Step 2

Next step varies between the paper and the appendix

- For each observation in $h_i^{balanced}$ the covariate values for the balancing variables W_i are compared with those of all other observations that share the same treatment/moderator assignment from the original data.
- Using nearest neighbors (Mahalanobis distance metric) each observation is then matched to the nearest observation in the original data.
- Replace the covariate values from the observation in $h_i^{balanced}$ with the matched observation from the original data.

Step 2

Next step varies between the paper and the appendix

- For each observation in $h_i^{balanced}$ the covariate values for the balancing variables W_i are compared with those of all other observations that share the same treatment/moderator assignment from the original data.
- Using nearest neighbors (Mahalanobis distance metric) each observation is then matched to the nearest observation in the original data.
- Replace the covariate values from the observation in $h_i^{balanced}$ with the matched observation from the original data.

No theoretical guarantees